

Vitamin Deficiency Risk Dataset Analytical Report

An exploratory overview for stakeholder review

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1. Dataset Overview

This dataset contains health, lifestyle, and nutritional information collected from individuals to support the development of a predictive model for vitamin deficiency risk. It captures a broad range of factors — from demographic characteristics and dietary habits to measured vitamin intake levels — alongside a continuous score representing overall deficiency risk.

Metric	Value
Dataset Name	train_data.csv
Total Records (Rows)	3,500
Total Features (Columns)	21
Numerical Columns	12
Categorical Columns	9
Target Variable	vitamin_deficiency (continuous score 0–1.89)

Column Descriptions

The table below lists every column in the dataset along with its data type and a plain-English explanation of what it measures.

Column Name	Data Type	Description
age	Numeric	Age of the individual in years (18–84)
gender	Category	Biological sex of the individual (Male / Female)
bmi	Numeric	Body Mass Index — a weight-to-height ratio used as a proxy for body fatness
smoking_status	Category	Whether the individual currently smokes, used to smoke, or has never smoked
alcohol_consumption	Category	Self-reported alcohol use level (Moderate / Heavy); 32% of rows are missing
exercise_level	Category	Physical activity level ranging from Sedentary to Active
diet_type	Category	Primary dietary pattern (Omnivore, Vegetarian, Vegan, Pescatarian)
sun_exposure	Category	Level of daily sun exposure (Low / Moderate / High), relevant for Vitamin D
income_level	Category	Household income bracket (Low / Middle / High)
latitude_region	Category	Geographic latitude band (Low / Mid / High) — affects sun availability
vitamin_a_percent_rda	Numeric	Daily Vitamin A intake expressed as a percentage of the Recommended Dietary Allowance
vitamin_c_percent_rda	Numeric	Daily Vitamin C intake as % of RDA
vitamin_d_percent_rda	Numeric	Daily Vitamin D intake as % of RDA — the most deficient nutrient in this dataset
vitamin_e_percent_rda	Numeric	Daily Vitamin E intake as % of RDA

vitamin_b12_percent_rda	Numeric	Daily Vitamin B12 intake as % of RDA — second most deficient nutrient
folate_percent_rda	Numeric	Daily Folate (Vitamin B9) intake as % of RDA
calcium_percent_rda	Numeric	Daily Calcium intake as % of RDA
iron_percent_rda	Numeric	Daily Iron intake as % of RDA
symptoms_count	Numeric	Number of deficiency-related symptoms reported by the individual (0–8)
symptoms_list	Category	Semicolon-separated list of specific symptoms (33% missing — indicates no symptoms)
vitamin_deficiency	Numeric	Target variable: composite deficiency risk score (0 = no deficiency, ~1.89 = highest risk)

2. Summary Statistics

2a. Numerical Columns

The table below summarises the distribution of every numerical column. Values have been rounded to two decimal places for readability.

Column	Mean	Median	Std Dev	Min	Max
Age	50.59	50.0	19.28	18.0	84.0
BMI	26.1	26.2	4.94	15.0	43.9
Vit A % RDA	90.68	85.5	37.15	10.0	219.0
Vit C % RDA	88.57	83.4	37.19	10.0	230.5
Vit D % RDA	72.42	62.3	42.99	7.0	275.6
Vit E % RDA	89.68	84.0	37.22	10.0	237.6
Vit B12 % RDA	62.7	55.9	37.44	10.0	243.6
Folate % RDA	90.2	84.6	37.28	10.0	226.6
Calcium % RDA	82.36	76.7	36.06	10.0	232.7
Iron % RDA	76.42	71.55	32.9	10.0	211.4
Symptom Count	1.87	2.0	1.79	0.0	8.0
Deficiency Score	0.65	0.66	0.45	0.0	1.89

2b. Categorical Columns

For each categorical column, the table below shows the number of distinct values and the most common response.

Column	Unique Values	Most Frequent Value / Category
gender	2	Male (1,781 — 50.9%)
smoking_status	3	Current (1,172 — 33.5%)
alcohol_consumption	2	Moderate (1,202 — 49.8% of non-missing)
exercise_level	4	Sedentary (903 — 25.8%)
diet_type	4	Vegan (887 — 25.3%)
sun_exposure	3	Moderate (1,187 — 33.9%)
income_level	3	Low (1,187 — 33.9%)
latitude_region	3	High (1,187 — 33.9%)
symptoms_list	228	228 unique symptom combinations

Notable Patterns from Summary Statistics

- Vitamin D has the lowest average intake (72% of RDA) and the highest variability — pointing to a widespread shortfall across the population.
- Vitamin B12 is the second most deficient nutrient, averaging only 63% of RDA. This is clinically significant as B12 deficiency can lead to neurological complications.
- The deficiency score ranges from 0 to 1.89 with an average of 0.65, suggesting that the majority of individuals carry some degree of nutritional risk.
- Nearly one-third of individuals (1,152 of 3,500) report zero symptoms, yet may still show nutritional shortfalls — highlighting the "silent" nature of early deficiencies.

3. Data Quality Assessment

3a. Missing Values

Two columns contain missing values. The table below shows the count and percentage of missing entries for each.

Column	Missing Rows	% Missing	Likely Explanation
alcohol_consumption	1,110	31.7%	Respondents may have declined to answer; consider imputation or separate "Unknown" category
symptoms_list	1,152	32.9%	Consistent with symptoms_count = 0 for the same rows; missing likely means "no symptoms reported"

Important: The 31.7% missing rate in alcohol_consumption is substantial. Before modelling, it is recommended to either (a) impute values using a suitable method, or (b) treat "Unknown" as a distinct category to avoid inadvertently discarding a third of the dataset.

3b. Duplicate Rows

A full scan of the dataset found **0 duplicate rows**. Every record is unique, which is a positive indicator of data collection integrity.

3c. Potential Anomalies & Data Quality Flags

Column / Area	Observation	Recommended Action
iron_percent_rda	Maximum value of 211% of RDA — more than double the daily requirement	Verify whether these are genuine high-intake cases or data entry errors
vitamin_d_percent_rda	Maximum reaches 275% of RDA; very high given the low average (72%)	Flag extreme outliers (>200%) for review before modelling
vitamin_b12, vitamin_c, vitamin_e, folate, calcium	Several vitamins exceed 200–230% of RDA at the upper end	Investigate whether these reflect supplement use not captured elsewhere
symptoms_list	228 unique symptom combinations make this column hard to use as-is	Consider one-hot encoding individual symptoms for modelling
vitamin_deficiency (target)	Highly continuous with 2,193 unique values — not a simple binary label	Clarify whether a regression or a threshold-based classification approach is intended

4. Key Insights

The following five observations summarise the most important things a stakeholder should know about this dataset before any modelling or further analysis begins.

01 Vitamin D Deficiency Is the Most Prevalent Nutritional Gap

On average, individuals in this dataset are only meeting 72% of their recommended daily intake for Vitamin D — the lowest average across all eight nutrients studied. Given that Vitamin D is primarily synthesised through sun exposure, the combination of geographic latitude and sun exposure variables in this dataset will be critical predictors for any model. This pattern warrants particular attention for individuals in high-latitude regions with low sun exposure.

02 Vitamin B12 Is the Second Most Widespread Deficiency

With an average intake of just 63% of RDA, Vitamin B12 represents a significant nutritional risk in this population. B12 is found almost exclusively in animal products, which may explain its low coverage — notably, the dataset includes a large proportion of vegan and vegetarian individuals (about 50% combined). This nutrient-diet interaction is likely to be one of the strongest predictive signals in the data.

03 One in Three Individuals Reports No Symptoms — Yet Deficiency Risk Still Exists

Approximately 1,152 individuals (33%) have a symptoms count of zero and no symptoms listed. However, the deficiency score for these individuals is not necessarily zero, meaning the data captures "silent" deficiency — a medically important phenomenon where nutritional shortfalls exist without obvious clinical signs. Any model trained on symptoms alone would miss a large portion of at-risk individuals.

04 Missing Alcohol Data Affects Nearly a Third of Records

The alcohol_consumption column is missing for 1,110 rows (31.7%). This is the single largest data quality issue in the dataset. Since alcohol use can affect nutrient absorption and metabolism, this variable may be an important predictor. A deliberate strategy — such as treating missingness as an "Unknown" category or applying statistical imputation — is needed before training any model, to avoid silently dropping a third of the data.

05 The Dataset Is Well-Balanced Across Lifestyle and Demographic Groups

Gender (50.9% Male, 49.1% Female), smoking status, exercise level, diet type, income level, and geographic region are all close to evenly distributed across their respective categories. This balance is a strength: models trained on this data are unlikely to be biased toward any single demographic subgroup, and results should generalise reasonably well across the represented population.

This report was generated for internal stakeholder review. All statistics are computed from the raw training file (train_data.csv) prior to any preprocessing or feature engineering. Further analysis, including correlation studies and model-readiness assessment, should be conducted before proceeding to the modelling phase.